Jacobus Langner

Full-Stack Engineer · Developer productivity · APIs, messaging and platform · .NET Core, Node, Python · Cape Town



jacobus@hack-cove.com • +27 78 629 4044 • GitHub: madangryscientist • LinkedIn

Senior full-stack engineer with 18 years of production software. Alongside delivering features, I have spent the last ten years building the architecture, APIs and tooling that let other developers excel, finding where they struggle and fixing it at the source. Back end in C#/.NET Core, Node and Python; front end in TypeScript and React. On call for the systems I build.

Experience

Founder / Contract Engineer — Hack Cove

Apr 2026 – Present

Client: Cue (voice and telephony)

- Built the Python service that integrates the app with the voice provider.
- Built the realtime call, transfer and presence flows on top of it in TypeScript and React: inbound answer, cold and warm transfers, and in-flight transfer actions.

PixelSlash

- Built a visual-regression service on Cloudflare Workers, D1 and Queues, with a Rust/WebAssembly pixel-diff engine, queued vision review of changed regions, and an MCP server so coding agents can drive it.

pixelslash.dev · github.com/madangryscientist/pixelslash

Omnibus

- Built a multi-agent engineering system in Bun/TypeScript: a planner agent dispatches builder agents into isolated git worktrees, each runs the test gate and opens a pull request, and leases stop agents colliding.

Hack Cove UI

- Built a reusable GitHub Action that reviews UI pull requests in two layers, deterministic rules then an LLM review, for an open-source React design system on npm.

hack-cove.com/ui · github.com/madangryscientist/hackcove

Tech: Python, PostgreSQL, TypeScript, Node/Bun, Cloudflare Workers, D1, Queues, Drizzle, Rust/WebAssembly, React

Senior Engineer — Zappi

Apr 2021 – Apr 2026

Developer platform and Node services · front-end guild lead (final year)

- Owned the developer platform for a 60+ package TypeScript monorepo used by 80 engineers across multiple applications.
- Cut library builds from 120 s to 5 s (CommonJS to ESM-only) and E2E runs from 30 to 5 min (Cypress to Playwright); moved 1,126 tests from Jest to Vitest (35% faster) and added Nx build caching.
- Built a Node microservice that took events from Salesforce's messaging system and republished them on SQS/SNS for other services to consume.
- Owned the Node.js backend-for-frontend (BFF): its build, and the APIs that sped up page loads.
- Led the front-end guild in the final year, setting technical direction across teams; led the React 18 migration and the move from Redux to React Query and Zustand, and cut bundle size from 41 MB to 24 MB.

Tech: Node.js, Amazon SQS/SNS, TypeScript, Nx, Playwright, Vitest, Ruby (back-end codebase), React, Next.js

API Architect — FMI

Jun 2018 – Mar 2021

API and platform architecture for a life insurer moving off paper

- Architected 15 API services and the workflow engines for policy applications, claims and underwriting, in C# and TypeScript.
- Built the company's Kubernetes clusters from scratch on Azure, and moved releases from monthly FTP drops to continuous delivery.
- Designed role-based access control and POPI Act compliance for medical and financial data.
- Delivered the React and Next.js front ends on those APIs.

Tech: C#, .NET Core, PostgreSQL, MongoDB, Kubernetes (AKS), Docker, TypeScript, React, Next.js

Software Development Team Lead — Wonga

Aug 2015 – May 2018

Migration architecture and tooling · team lead

- Drove the architecture to move the business off an unsupported PHP site and NServiceBus, and was key in building the migration tooling used by three teams.
- Led a team of seven building new products as C# microservices with CQRS and RabbitMQ on ASP.NET Core Web API.
- Joined as a back-end developer, then became technical lead for the move from PHP to AngularJS on ASP.NET Core Web API. The team won the CEO Recognition Award.

Tech: C#, ASP.NET Core Web API, RabbitMQ, NServiceBus, CQRS, Octopus Deploy, AngularJS

Earlier Roles

Software Developer — Sygnia — Sep 2014 – Aug 2015
Linked the web and server-side teams in asset management; tuned slow queries.

Software Developer — Property24 — Aug 2011 – Aug 2014
Built and ran one of South Africa's most visited websites across multiple servers with heavy caching; upgraded it from ASP Classic to ASP.NET to ASP.NET MVC.

Software Developer — ACI Worldwide — Jan 2010 – Aug 2011
Java on Windows and AIX with SQL Server and DB2; third-level support and patches for the PostCard payments product.

Software Developer — Axxess DSL and Business Intelligence — Sep 2008 – Dec 2009
C# Windows services, web services and ASP.NET sites on SQL Server; upgrades from ASP.NET 1.1 to 3.5.

Education

Bachelor of Science (BSc), Computer Science — NMMU, Port Elizabeth

2005–2008 • Automata Theory 83% • Linear Optimization 88% • Advanced Programming 71%/76%

Skills

Back end: C#/.NET Core, ASP.NET Core Web API, Node.js, Python, REST APIs, microservices, event-driven architecture, CQRS

Messaging: RabbitMQ, NServiceBus, Amazon SQS/SNS, Cloudflare Queues

Databases: PostgreSQL, SQL Server, MongoDB, Cloudflare D1

Platform & DevEx: Nx monorepos, CI/CD, build and test performance, internal developer tooling, AI-agent tooling (MCP)

Infrastructure: Kubernetes (Azure AKS), Docker, AWS, Cloudflare Workers, GitHub Actions, GitLab CI, Octopus Deploy

Front end: TypeScript, React, Next.js, Angular, Vue

Build & testing: Vite, tsdown, Webpack, Vitest, Playwright, Storybook

Spoken: English (native), Afrikaans (fluent)